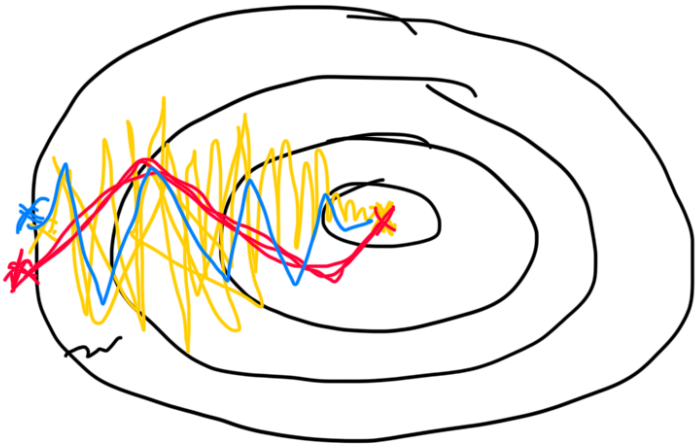
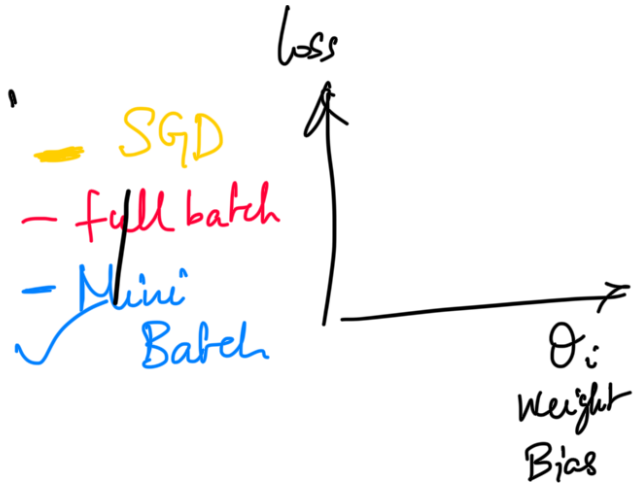


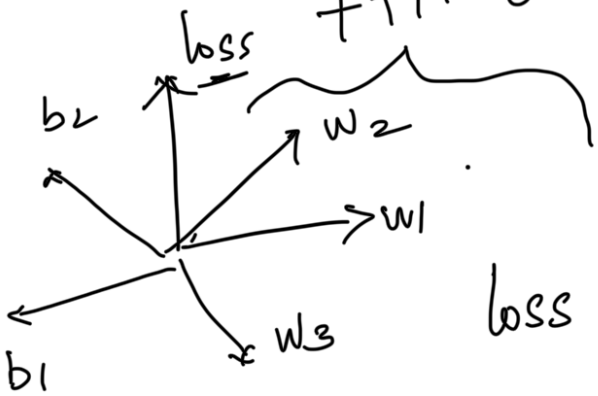


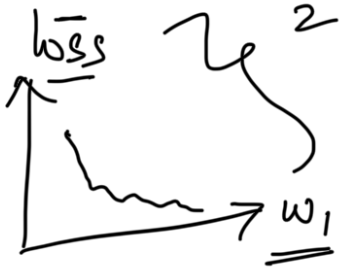
Fifteen.

$\theta = [w_1, w_2, w_3, w_4, B_1, B_2, B_3]$ ⇒ dimension of my loss surface.

$\frac{dJ}{dw_1}$ ⇒ Involving other parameters as well

7+1 = 8 dims.





$loss = \frac{1}{2}(y_i - \hat{y}_i)^2$ ⇒ one row

$cost = \frac{1}{N}\sum_{i=1}^{N} loss$ ⇒ all rows. Mini Batch

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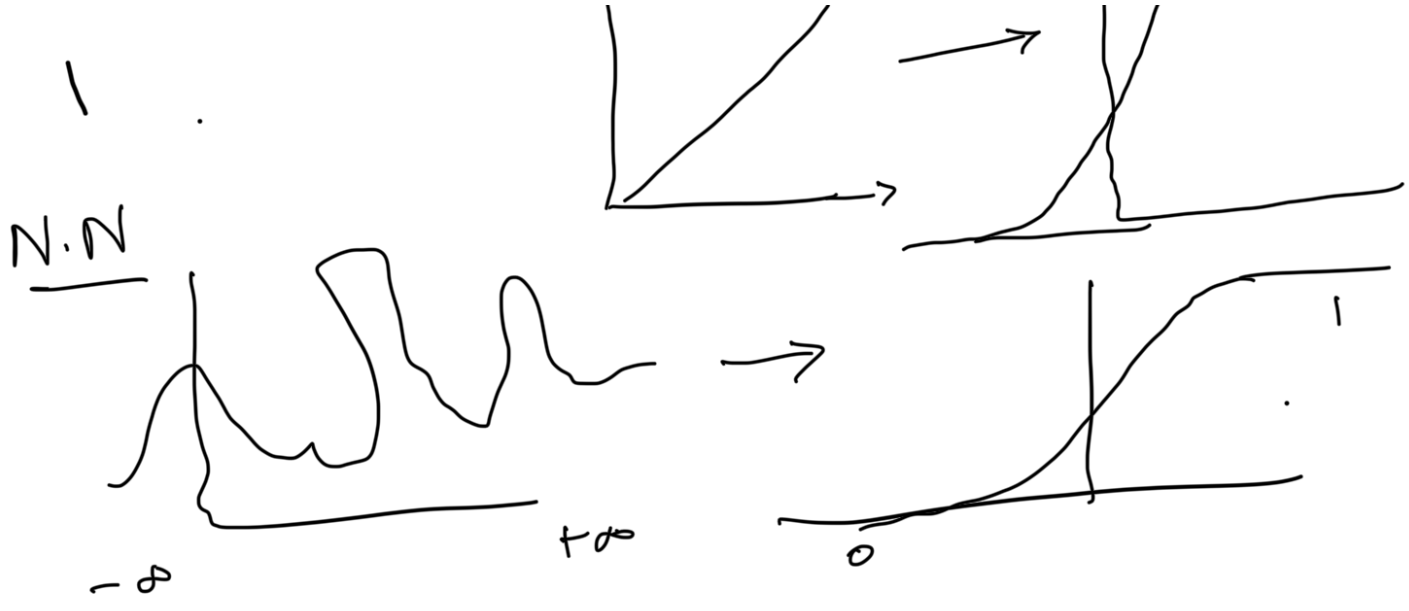
Sigmoid → o/p ⇒ 0–1 / Binary.

Softmax → o/p → [0, 1, 2, 3] / Multiclass

Continuous Values to probs

$x \to x$ / → linear ⇒ Continuous.

Task: logistic Reg.



	Sales	Ice/cm
Mon	50	
Tues	52	
~~Wed~~	49 ⇒	
~~Thur~~	80 ←	
~~Fri~~	51	
~~Sat~~	50	
~~Sun.~~	53	

AGENDA :- 0th W EMA

1. G.D with momentum
2. RMS Prop
3. ADAM.
4. MNIST Image Classification with NN

Next week. ⇒ 5. Deploy in AWS EC2

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① EWMA ⇒ Exponentially Weighted Moving Average.

Representation Learning → NN - experts in deep representations

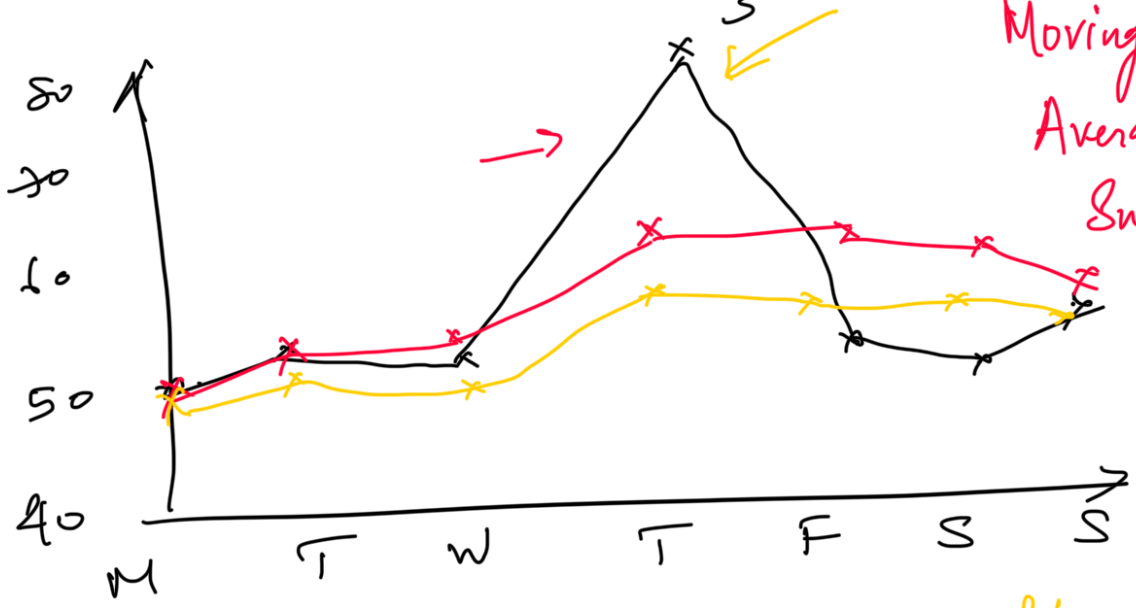
Avg 3 day sales for Wednesday

$= \frac{50+52+49}{3} = 50.33$

3

for Tuesday = $\frac{52+49+80}{3} = 60.33$

for Friday = $\frac{49+80+81}{3} = 60$



Moving Average Smoothening

EMA

⇒ I give weightage to days / More weightage to recent past & less weightage to older past

80% of my previous estimate & Incorporate 20% of current est.

EMA = 0.8(Previous EMA) + 0.2(Current Value)

M = 50

Thursday

Tuesday = 0.8(50) + 0.2(52) = 50.4

W = 0.8(50.4) + 0.2(49) = 50.12

T = 0.8(50.12) + 0.2(80) = 56.096

Actual ∞ | Smoothed value 56.1

SMOOTHENING

Part II What if I apply the same idea to noisy gradients calculated during training?



$\hat{y} = f(x, \theta)$ ⇒ ① forward pass ② loss ③ Back propagation ④ param update ⑤ Repeat.

EMA ⇒ parameter update not actual

$\theta = [w_1, b_1, w_2, b_2, w_3, w_4, b_3]$

$w_{new} = w_{old} - \eta g$

$\theta_{new} = \theta_{old} - \eta \frac{dJ \Rightarrow \text{cost}}{d\theta \Rightarrow \theta}$ | $\frac{dJ}{d\theta} = g$

$\theta_{new} = \theta_{old} - \eta g \Rightarrow$ just depends on

learning rate is also constant

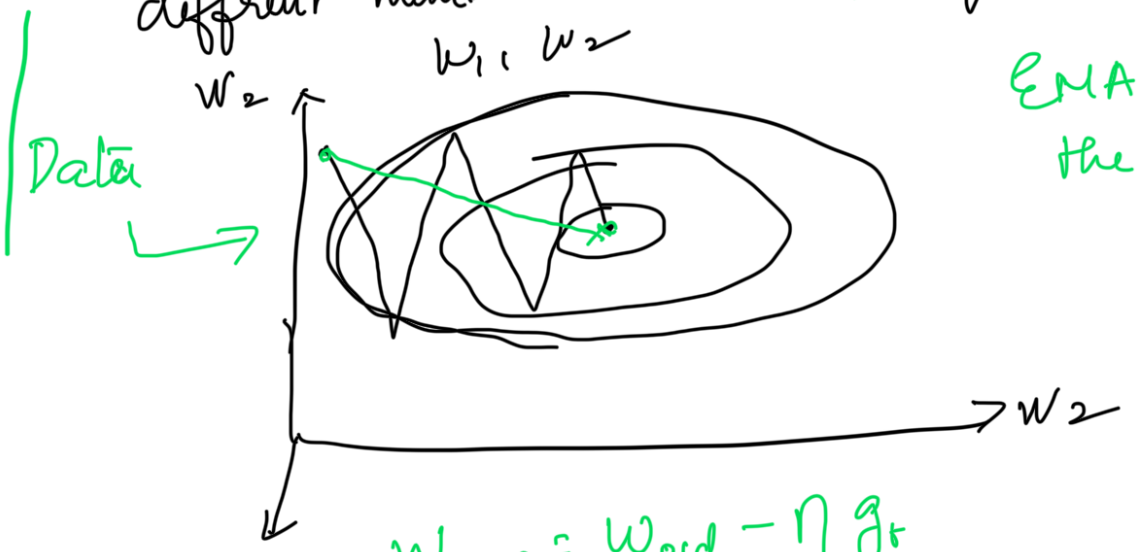
Current value of dJ/dw

Prob: g_t is based on only current value & is not intelligent.

Big ⇒ low ↑

w_1 , w_2

different mini-batches Calculate gradies for both



EMA over the Gradients

$$w_{new} = w_{old} - \eta g_t$$

EMA of gradient

$$m_t = \beta m_{t-1} + (1-\beta) g_t$$

(g_t → current)

Previous EWMA

$$w_{new} = w_{old} - \eta m_t$$

⇒ Intelligent

Previous gradient aware new gradient EWMA ↗

Gradient descent with momentum.

S.G.D with momen.

$$W_{new} = W_{old} - \eta \underbrace{g_t}_{\text{current grad}}$$

With momentum:

$$W_{new} = W_{old} - \eta \underbrace{m_t}_{\text{Gradient history}}$$

Part III

We now have a way to smooth inconsistent gradients. But what if the gradients for one parameter are usually very large, while gradients for other parameters are small?

Momentum helps with capturing consistent direction.

Now the step size has to be co-ordinated for each parameter.

RMSprop:

$$W_{new} = W_{old} - \eta g_t$$

$$S_t = \rho S_{t-1} + (1-\rho) g_t^2$$

↙ Prev. Moving avg.

⇒ Removes -ve & just keeps magnitude

$$W_{new} = W_{old} + \frac{\eta}{\sqrt{S_t + \epsilon}} g_t$$

↑ to avoid ÷ by '0'

Curr ↑ Prev ↓ Step

↓, ↑

Drop − η

Root Mean Square

L.R nottobe fixed I will have Adaptive Step.

2 Ideas

Momentum
EMA of Signed gradients

RMS prop
EMA of Squared grads
Scaling η

IV ADAM

Combined

$$W_{new} = W_{old} - \frac{\eta}{\sqrt{V_t + \varepsilon}} m_t$$

→ Momentum

RMS prop ← $\sqrt{V_t + \varepsilon}$ → Small number to avoid it by 0